

Mixed effects models

for analysis of repeated measurements

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First question on advising:

Is the question really repeated measurements
or just change from start to end of study?

Do we always need mixed effects models?

- When we compare first and last observation;

Why not just work with difference last-first?

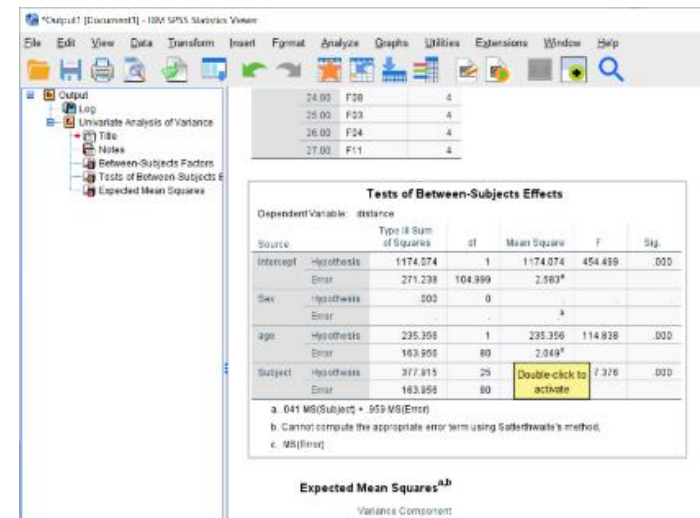
(paired samples t-test & confidence intervals)

- One option to first take the mean of each group?
(for balanced designs without missing)

How do we analyze repeated measurements?

ANOVA (analysis of variance)

- Old classical approach
- Easy to calculate by hand
- ... but:
 - > Gives p-values, not estimated effects with confidence interval
 - > No unbalanced missing data
 - > Not very flexible



The screenshot shows the SPSS Statistics Viewer interface. The left pane displays a tree view of the output, including 'Univariate Analysis of Variance'. The right pane shows the 'Tests of Between-Subjects Effects' table for the dependent variable 'distance'.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	1174.074	1	1174.074	454.489	.000
Error	271.238	104.999	2.583 ^a		
Sex	.000	0	.	.	
Age	235.356	1	235.356	114.838	.000
Subject	163.856	80	2.049 ^b		
Error	163.856	80	2.049 ^b		
Subject	277.815	25	11.113	7.376	.000
Error	163.856	80	2.049 ^b		

Double-click to activate

a. 041 MS(Subject) = 959 MS(Error)
b. Cannot compute the appropriate error term using Satterthwaite's method.
c. MS(Error)

Expected Mean Squares^{a,b}

Variance Component

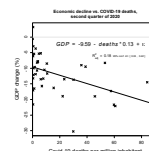


An endangered method for repeated measurements?

How do we analyze repeated measurements?

Mixed models regression

- The modern flexible approach to repeated measurements
- *Works in:*
R, Stata, Python, jamovi, Julia, and SPSS
- ***An extension of ordinary regression analysis***
- *Estimates effects with confidence intervals*



Random effects (Modelling individual effects)

Classic basic statistical model: (linear regression etc.)

$$Y = \text{“covariate effects”} + \text{“random noise”}$$

A mixed effects model with random effects:

$$Y = \text{“covariate effects”} + \text{“individual variation”} + \text{“random noise”}$$

=> We take into account individual variation by random effects
(random effects might be used for all sorts of cluster data)

Random effects (Modelling individual effects)

Classic basic statistical model: (linear regression etc.)

$$Y_i = \alpha + \beta * X + \varepsilon_i$$

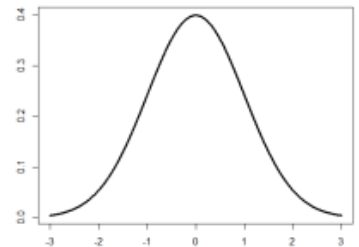
A mixed effects model with random effects:

$$Y_{i,j} = \alpha + \beta * X + \mathbf{u}_{j.} + \varepsilon_{i,j}$$

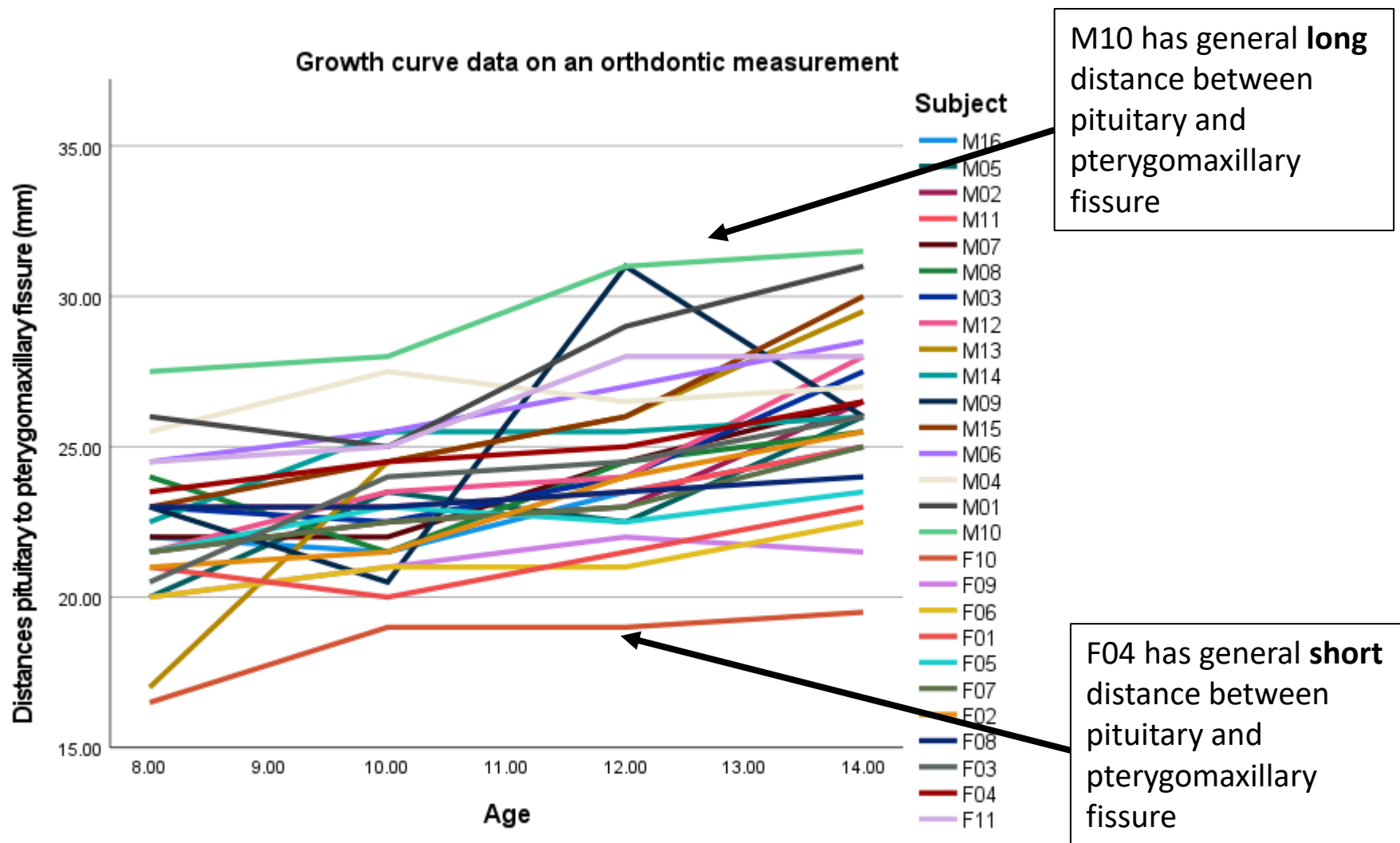
Assumptions;

The individual level comes from a normal distribution,

(like the normal distributed random noise)



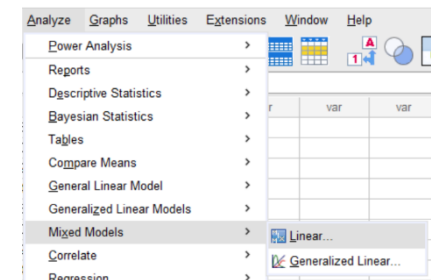
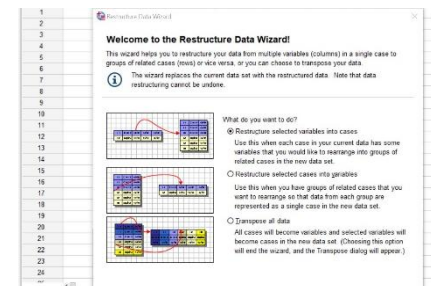
Random (individual) effects



Data format for mixed models

- Mixed effects models take data on “long format”;
One observation per line with shared id.
- Stata/R/Python/SPSS have routines to convert from “wide” to “long” format
- Stata/R/Python/SPSS all have comprehensive mixed models estimation routines

distance	age	Subject	Sex
22.50	8.00	M14	Male
25.50	10.00	M14	Male
25.50	12.00	M14	Male
26.00	14.00	M14	Male
23.00	8.00	M15	Male
24.50	10.00	M15	Male
26.00	12.00	M15	Male
30.00	14.00	M15	Male
22.00	8.00	M16	Male
21.50	10.00	M16	Male
23.50	12.00	M16	Male
25.00	14.00	M16	Male
21.00	8.00	F01	Female
20.00	10.00	F01	Female
21.50	12.00	F01	Female
23.00	14.00	F01	Female



Mixed effects models could be both:

- Linear regression
- Logistic regression
- Poisson regression
- Survival analysis ("frailty models")

Remark: Logistic regression can be unstable when limited data is available

Mixed effects models might be complex

- Many different levels/clusters
- Advanced covariance and dependent structures
- Non-linear expected (mean) values
- With added smoothing splines etc.

Mixed effects models - Intuition

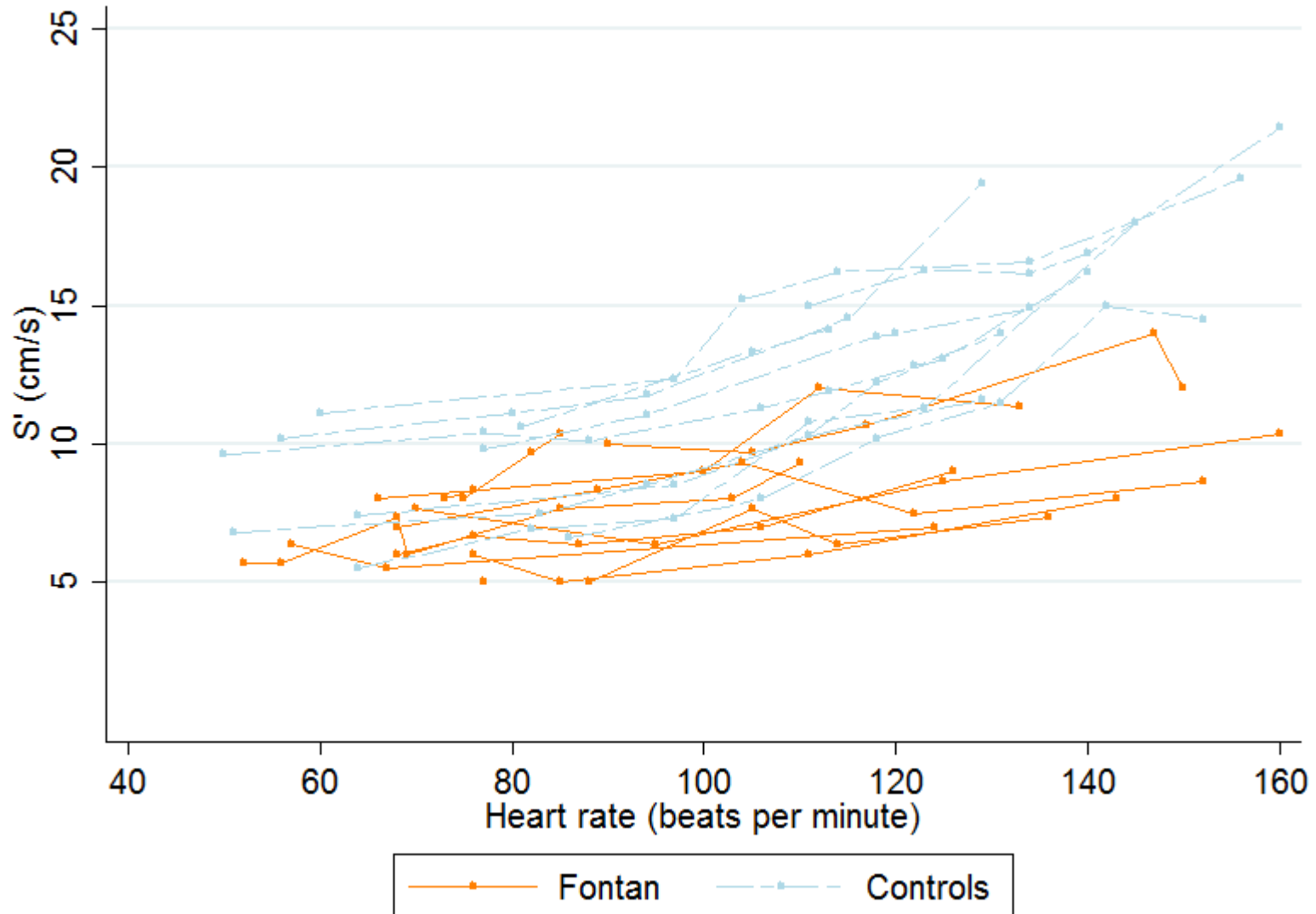
Individual with more observations influence the estimates most, but adding new individuals is usually more important than one more repeated observation

Prediction with mixed effects models

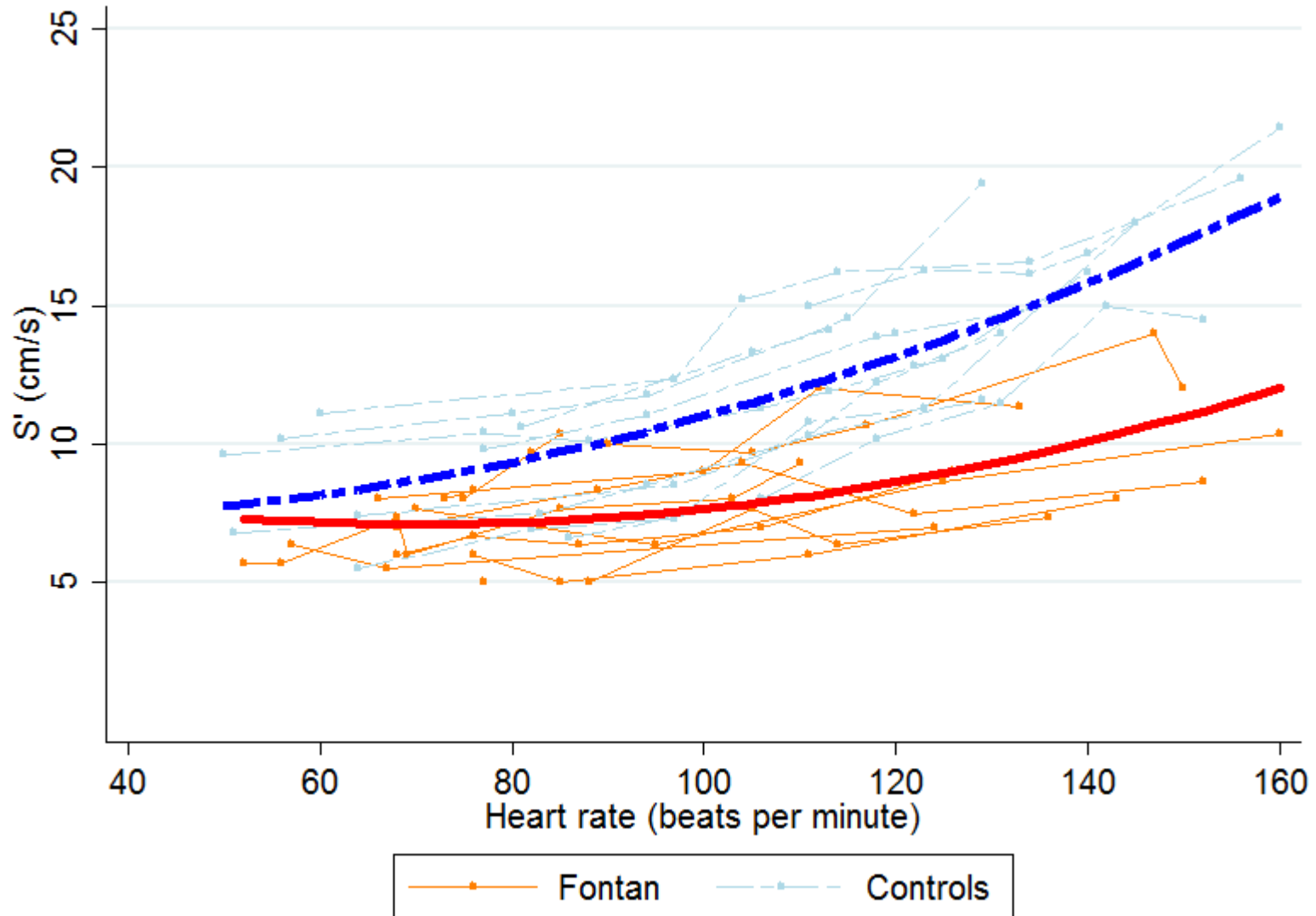
- Two distinct prediction goals:
 - > **Marginal (population-average):** Sets random effects to zero
 - > **Conditional (subject-specific):** Uses empirical Bayes estimates for Best Linear Unbiased Predictor [BLUP]. Can be used for personalized prognostic predictions.

Mark that *mixed effects models* is not built on imputation,
but estimated by maximum likelihood (or REML)
integrating over the random effects.

Cardiac performance during exercise with Fontan circulation - data



Cardiac performance during exercise with Fontan circulation - data



Model diagnostics for mixed effects models

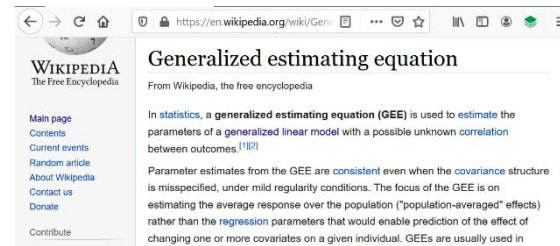
- **Structure:**
Linear relationships, possible individual random slopes, overdispersion (typically relevant for Poisson regression) etc.
- **Residuals (ϵ):**
Homoscedasticity, normality, spatial/temporal independence.
- **Random Effects (b):** Normal distribution, no outliers, uncorrelated with Level-1 predictors.

PS: Take software warnings seriously

Non normal distributed data? (continuous variables)

- **Generalized estimating equations (gee)**

An alternative for
non normal distributed data
with decent number of individuals



- **Bootstrap**

a good robust method,
but be careful with SPSS
(standard implementation not recommended!)



Summary:

- 1. Look first for simplifications** (marginal means etc.)
- 2. Use mixed effects models with random components for complex data taking into account individual variations**

Remark:

Mixed effects models is an extension of ordinary generalized linear models with individual variation (but can become very complex)

Questions



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Vcard

See practical exercises at:

https://github.com/ocbe-uio/public-slides/tree/main/Mini_courses/MixedModels_mini_course/

«MixedModels_mini_course_slides.pdf»	Intro to mixed models
«R_mixed_models_quicksheet.html»	Mixed models in R quick sheet
«Sleepdata_R_exercise.html»	A small exercise
«data\sleepstudy_wide.csv»	Example data for exercise
«Sleepdata_R_exercise_results.ipynb»	Solution to exercise
«Sleepdata_R_exercise_results.html»	Solution to exercise in html



Referanser:

- **Mixed models:**

- > https://en.wikipedia.org/wiki/Mixed_model
- > <https://stats.idre.ucla.edu/other/mult-pkg/introduction-to-linear-mixed-models/>

- **Software for mixed effects models:**

- > R:
<https://www.r-project.org/>
<https://cran.r-project.org/web/packages/lme4/vignettes/lmer.pdf>
- > Python:
<https://www.python.org/>
https://www.statsmodels.org/stable/mixed_linear.html
- > Stata:
<https://www.stata.com/>
<https://www.stata.com/features/multilevel-mixed-effects-models/>
- > SPSS:
<https://www.ibm.com/analytics/spss-statistics-software/>
<https://stats.idre.ucla.edu/spss/seminars/spss-mixed-command/>

R code:

```
> library(nlme)
> orthodont <- read.csv("eks-data/Orthodont.csv")
> est <- lme(distance~sex+age, random=~age | subject, data=orthodont)
> summary(est)
```

```
Linear mixed-effects model fit by REML
Data: orthodont
      AIC      BIC    loglik
449.2339 467.8116 -217.6169

Random effects:
Formula: ~age | Subject
Structure: General positive-definite, Log-Cholesky parametrization
          StdDev   Corr
(Intercept) 2.7970227 (Intr)
age         0.2264274 -0.766
Residual    1.3100398

Fixed effects: distance ~ Sex + age
              Value Std.Error DF   t-value p-value
(Intercept) 17.635200 0.8862449  80 19.898788   0.000
Sex         -2.145492 0.7574539  25 -2.832504   0.009
age          0.660185 0.0712532  80  9.265338   0.000
Correlation:
      (Intr) Sex
Sex    -0.348
age   -0.838  0.000

Standardized Within-Group Residuals:
          Min           Q1           Med            Q3           Max
-3.08141704 -0.45675583  0.01552695  0.44704158  3.89437694

Number of Observations: 108
Number of Groups: 27
```

```
> VarCorr(est)$subject
```

```
      (Intercept)      age
(Intercept)  7.8236561 -0.48503538
age         -0.4850354  0.05126868
attr(,"stddev")
      (Intercept)      age
2.7970799  0.2264259
attr(,"correlation")
      (Intercept)      age
(Intercept)  1.0000000 -0.7658478
age         -0.7658478  1.0000000
```

R code with lme4 package:

```
> library(lme4)          # Run first:  install.packages("lme4")
> orthodont <- read.csv("eks-data/Orthodont.csv")

> est <- lmer(distance ~ sex+age+(1+age|subject), data=orthodont)
> summary(est)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: distance ~ sex + age + (1 + age | subject)
Data: orthodont
```

```
REML criterion at convergence: 435.2
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-3.0821	-0.4568	0.0154	0.4471	3.8941

```
Random effects:
```

Groups	Name	Variance	Std.Dev.	Corr
subject	(Intercept)	7.81591	2.7957	
	age	0.05123	0.2263	-0.77
Residual		1.71644	1.3101	

Number of obs: 108, groups: subject, 27

```
Fixed effects:
```

	Estimate	Std. Error	t value
(Intercept)	17.63510	0.88612	19.902
sex	-2.14524	0.75743	-2.832
age	0.66019	0.07125	9.266

```
Correlation of Fixed Effects:
```

	(Intr)	sex
sex	-0.348	
age	-0.838	0.000

```
> VarCorr(est)$subject
```

```
(Intercept)      age
(Intercept)  7.8159130 -0.48446528
age          -0.4844653  0.05122676
attr(,"stddev")
(Intercept)      age
2.7956954  0.2263333
attr(,"correlation")
(Intercept)      age
(Intercept)  1.0000000 -0.7656395
age          -0.7656395  1.0000000
```

R wide to long format:

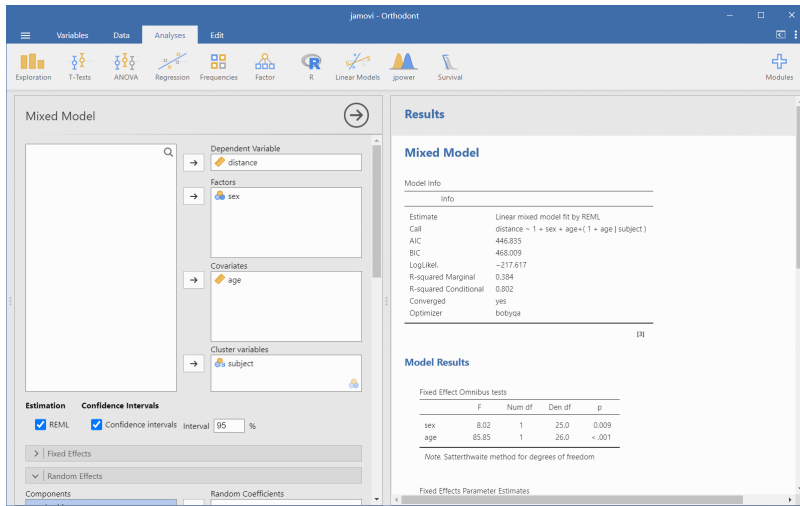
```
# Sample Wide Data:
```

```
> df_wide <- data.frame(  
  id      = 1:3,  
  visit_1 = c(12, 14, 11),  
  visit_2 = c(15, 17, 13),  
  visit_3 = c(19, 22, 16)  
)
```

```
# Wide to long using tidyr:
```

```
library(tidyr)  
df_long <- pivot_longer(  
  df_wide,  
  cols      = starts_with("visit_"), # columns to pivot  
  names_to  = "visit",               # new column name  
  names_prefix = "visit_",           # drop "visit_" prefix  
  names_transform = as.numeric,      # convert to numeric (1, 2, 3)  
  values_to  = "score"               # name of new value column  
)
```

R jamovi GUI:



The screenshot shows the R jamovi GUI with the 'Mixed Model' analysis selected. The 'Dependent Variable' is 'distance', and the 'Factors' are 'sex' and 'age'. The 'Cluster variables' are 'subject'. The 'Estimation' section shows 'REML' and 'Confidence intervals' checked, with an interval of 95%. The 'Results' panel displays the following information:

Mixed Model

Model info

Estimate	Linear mixed model fit by REML
Call	distance ~ 1 + sex + age + (1 + age subject)
AIC	446.835
BIC	468.009
LogLik	-217.617
R-squared Marginal	0.384
R-squared Conditional	0.802
Converged	yes
Optimizer	bobyqa

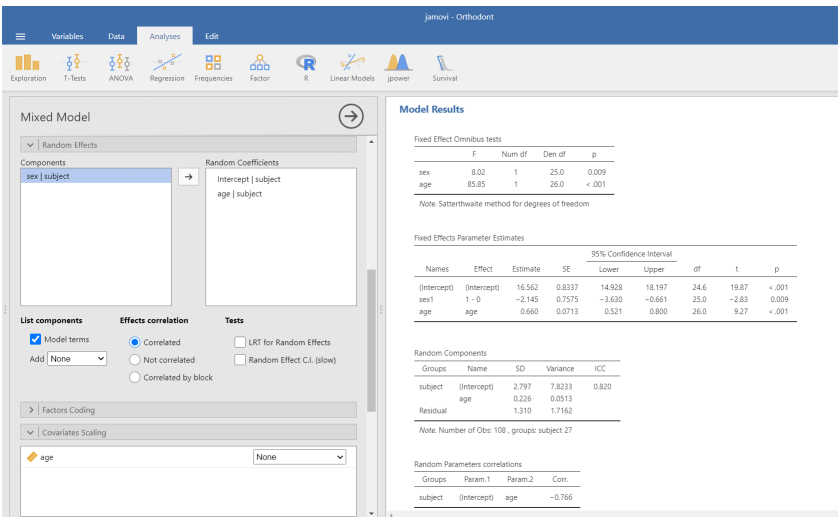
Model Results

Fixed Effect Omnibus tests

	F	Num df	Den df	p
sex	8.02	1	25.0	0.009
age	85.85	1	26.0	< .001

Note: Satterthwaite method for degrees of freedom

Fixed Effects Parameter Estimates



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Model Results

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	F	Num df	Den df	p
sex	8.02	1	25.0	0.009
age	85.85	1	26.0	< .001

Note: Satterthwaite method for degrees of freedom

Fixed Effects Parameter Estimates

Names	Effect	Estimate	SE	95% Confidence Interval	df	t	p	
(Intercept)	(Intercept)	16.562	0.8337	14.908	18.197	24.6	19.87	< .001
sex	1 + 0	-2.145	0.7375	-3.630	-0.661	25.0	-2.83	0.009
age	age	0.660	0.0713	0.521	0.800	26.0	9.27	< .001

Random Components

Groups	Name	SD	Variance	ICC
subject	(Intercept)	2.797	7.8233	0.820
	age	0.226	0.0513	
Residual		1.310	1.7162	

Note: Number of Obs: 108, groups: subject 27

Random Parameters correlations

Groups	Param.1	Param.2	Corr.
subject	(Intercept)	age	-0.766

Python code:

```
> import pandas as pd
> import statsmodels.formula.api as smf

> df = pd.read_csv("eks-data/Orthodont.csv", sep=", ")

> m = "distance~sex+age"
> est = smf.mixedlm(m, data=df, groups=df["subject"], re_formula="~age")
> est.fit().summary()
```

```
=====
Model:                MixedLM   Dependent Variable:  distance
No. Observations:    108        Method:              REML
No. Groups:          27         Scale:            1.7162
Min. group size:     4          Log-Likelihood:     -217.6169
Max. group size:     4          Converged:         Yes
Mean group size:     4.0

-----
              Coef.  Std.Err.   z      P>|z| [0.025 0.975]
-----
Intercept      17.635    0.897  19.656  0.000  15.877  19.394
sex            -2.145    0.831  -2.581  0.010  -3.775  -0.516
age             0.660    0.071   9.265  0.000   0.521   0.800
Group Var       7.823    4.910
Group x age Cov -0.485    0.387
age Var         0.051    0.035
=====
```

Stata code:

```
. import delimited "eks-data\Orthodont.csv"

. mixed distance age sex || subject: age, covariance(unstructured) reml
```

Performing EM optimization ...

Performing gradient-based optimization:

```
Iteration 0:  log restricted-likelihood = -217.83538
Iteration 1:  log restricted-likelihood = -217.77889
Iteration 2:  log restricted-likelihood = -217.61729
Iteration 3:  log restricted-likelihood = -217.61693
Iteration 4:  log restricted-likelihood = -217.61693
```

Computing standard errors ...

```
Mixed-effects REML regression           Number of obs   =       108
Group variable: subject                 Number of groups  =        27
                                         Obs per group:
                                         min =         4
                                         avg =        4.0
                                         max =         4
                                         Wald chi2(2)     =       93.87
                                         Prob > chi2      =      0.0000

Log restricted-likelihood = -217.61693
```

	distance	Coefficient	Std. err.	z	P> z	[95% conf. interval]
age		.6601852	.0712533	9.27	0.000	.5205313 .799839
sex		-2.145489	.7574537	-2.83	0.005	-3.630071 -.6609069
_cons		17.6352	.8862448	19.90	0.000	15.89819 19.37221

Random-effects parameters		Estimate	Std. err.	[95% conf. interval]
subject: Unstructured				
var(age)		.0512696	.0414506	.0105118 .2500589
var(_cons)		7.823338	5.730703	1.861552 32.87827
cov(age, _cons)		-.4850295	.4624058	-1.391328 .4212692
var(Residual)				
		1.716203	.3302832	1.176942 2.502548

```
LR test vs. linear model: chi2(3) = 49.29          Prob > chi2 = 0.0000
```

Note: LR test is conservative and provided only for reference.

Example data:



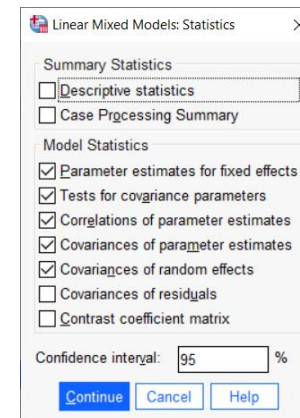
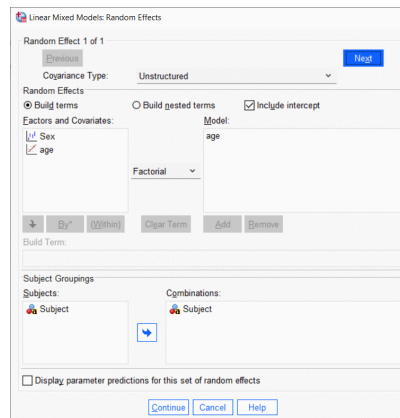
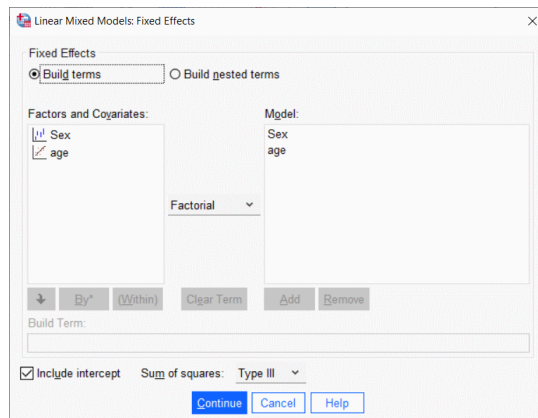
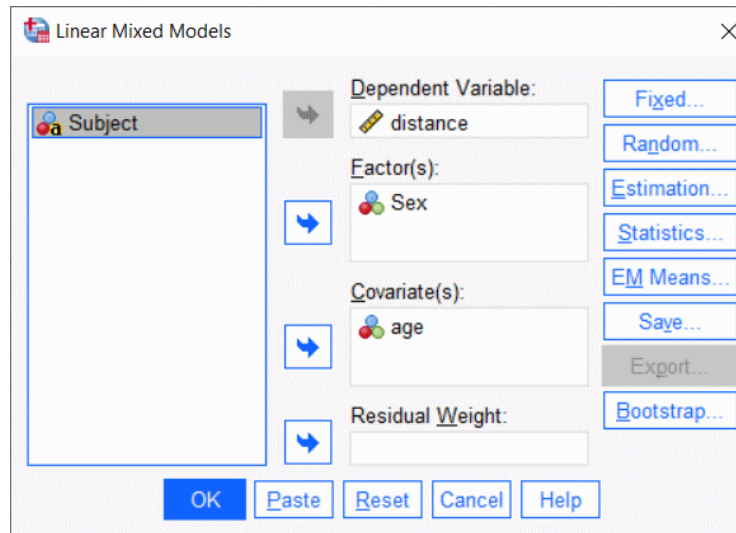
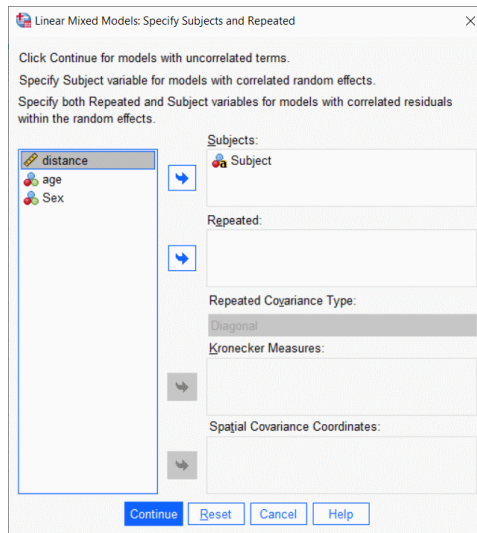
Orthodont.csv

SPSS code:

```
GET DATA  /TYPE=TXT
  /FILE="eks-data\Orthodont.csv"
  /ENCODING='UTF8'
  /DELIMITERS=","
  /FIRSTCASE=2
  /VARIABLES=
distance AUTO
age AUTO
subject AUTO
sex AUTO
/MAP.
EXECUTE.

MIXED distance BY sex WITH age
  /CRITERIA=DFMETHOD(SATTERTHWAITE) CIN(95) MXITER(100) MXSTEP(10) SCORING(1)
  SINGULAR(0.000000000001) HCONVERGE(0.00000001, RELATIVE)
  LCONVERGE(0, ABSOLUTE) PCONVERGE(0,ABSOLUTE)
  /FIXED=Sex age | SSTYPE(3)
  /METHOD=REML
  /PRINT=CORB COVB G  SOLUTION TESTCOV
  /RANDOM=INTERCEPT age | SUBJECT(subject) COVTYPE(UN) .
```

SPSS menus (GUI)



SPSS output:



Adobe Acrobat
Document

Julia:

```
> using CSV
> df = CSV.File("eks-data/Orthodont.csv", delim=',')

> using MixedModels
> formula = @formula(distance ~ sex+age + (1 +age | subject))
> est = fit(MixedModel, formula, df, REML=true)
```

```
Minimizing 136    Time: 0:00:00 ( 6.51 ms/it)
Linear mixed model fit by REML
  distance ~ 1 + sex + age + (1 + age | subject)
  REML criterion at convergence: 435.23385746662103
```

```
Variance components:
      Column  Variance Std.Dev.   Corr.
subject (Intercept)  7.824112 2.797161
      age           0.051273 0.226435 -0.77
Residual           1.716183 1.310032
Number of obs: 108; levels of grouping factors: 27
```

Fixed-effects parameters:

	Coef.	Std. Error	z	Pr(> z)
(Intercept)	17.6352	0.886259	19.90	<1e-87
sex	-2.14553	0.757457	-2.83	0.0046
age	0.660185	0.0712539	9.27	<1e-19